

Lecture 14: Entropy Balance for Open Systems

CBE 20260 / Thermodynamics I

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1 Introduction

In the previous lecture, we developed the entropy balance for **closed systems** (fixed mass). Today, we extend this to **Open Systems**, where mass flows across the system boundaries.

This allows us to analyze continuous engineering devices like turbines, compressors, heat exchangers, and throttle valves.

2 The General Entropy Balance

Just like the First Law, the Second Law for an open system accounts for:

1. Rate of change of entropy within the system.
2. Entropy carried in/out by mass flow.
3. Entropy transfer via heat (recall: Work does not directly transfer entropy).
4. Entropy **Generation** (the irreversible part).

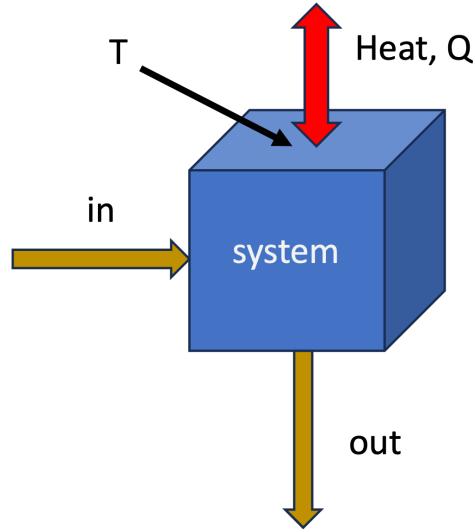


Fig. 1: System we apply the open system entropy balance to.

The general rate form, applied to the system shown in Fig. 1 is:

$$\frac{d(M\hat{S})}{dt} = \sum_{in} \dot{m}_{in} \hat{S}_{in} - \sum_{out} \dot{m}_{out} \hat{S}_{out} + \sum_j \frac{\dot{Q}_j}{T_j} + \dot{S}_{gen} \quad (1)$$

where

- $\dot{m}\hat{S}$: The convective transport of entropy (entropy moving with fluid).
- $\dot{Q}/T$: Entropy transfer accompanying heat transfer at boundary temperature T . There could be multiple heat transfer paths, hence the sum over j .
- $\dot{S}_{gen}$: Rate of entropy generation due to irreversibilities. $\dot{S}_{gen} \geq 0$.

With the exception of the generation term, this is similar to the First Law statement of **energy conservation**. Entropy has a generation term, however, that accounts for the fact that real processes are irreversible and generate entropy. We do not have entropy conservation in the same way that we have energy conservation.

The integrated form of Eq. 1 for a finite time interval is:

$$\Delta S = \sum_{in} m\hat{S} - \sum_{out} m\hat{S} + \sum_{heat} \frac{Q}{T} + S_{gen} \quad (2)$$

where m represents the mass that flowed in or out during the process. Indices have been dropped for simplicity.

2.1 Special Case: Steady-State

For steady-state processes (most continuous operations), no properties change with time inside the system so we have

$$\frac{d(M\hat{S})}{dt} = 0 \text{ or } \Delta S = 0.$$

Time-dependent form

$$0 = \sum_{in} \dot{m}_{in} \hat{S}_{in} - \sum_{out} \dot{m}_{out} \hat{S}_{out} + \sum_j \frac{\dot{Q}_j}{T_j} + \dot{S}_{gen}$$

Integrated form

$$0 = \sum_{in} m_{in} \hat{S}_{in} - \sum_{out} m_{out} \hat{S}_{out} + \sum_j \frac{Q_j}{T_j} + S_{gen}$$

Rearranging to solve for generation:

$$\dot{S}_{gen} = \sum_{out} \dot{m}_{out} \hat{S}_{out} - \sum_{in} \dot{m}_{in} \hat{S}_{in} - \sum_j \frac{\dot{Q}_j}{T_j} \geq 0$$

or

$$S_{gen} = \sum_{out} m_{out} \hat{S}_{out} - \sum_{in} m_{in} \hat{S}_{in} - \sum_j \frac{Q_j}{T_j} \geq 0$$

Single-Inlet, Single-Outlet (SISO)

For a device with one inlet and one outlet (like a pump or compressor) at steady state:

$$\dot{S}_{gen} = \dot{m}(\hat{S}_{out} - \hat{S}_{in}) - \frac{\dot{Q}}{T}$$

3 Example 1: Non-Reversible Compressor

Consider a steady-state air compressor.

- **Fluid:** Air (ideal gas, diatomic, $C_P \approx \frac{7}{2}R$).
- **System:** The compressor itself (the control volume includes the blades, casing, etc.).
- **Inlet (1):** 1 bar, 290 K.
- **Outlet (2):** 10 bar, 600 K.
- **Heat Transfer:** The compressor is adiabatic ($\dot{Q} = 0$).

- **Mass Flow:** $\dot{m} = 1 \text{ mol/s}$ (molar basis).

Goal: Calculate the work required and the entropy generation.

3.1 Step 1: First Law (Energy Balance)

Open system energy balance, steady-state, adiabatic, KE and PE negligible:

$$0 = \dot{m}(\hat{H}_{in} - \hat{H}_{out}) + \dot{W}_s$$

or

$$\Delta \dot{H} = \dot{W}_s$$

$$\dot{W}_s = \dot{m}(\hat{H}_{out} - \hat{H}_{in}) = \dot{m}C_P(T_2 - T_1)$$

$$\dot{W}_s = (1 \frac{\text{mol}}{\text{s}})(29.1 \frac{\text{J}}{\text{mol K}})(600 - 290 \text{ K})$$

$$\dot{W}_s = +9,021 \text{ Watts}$$

3.2 Step 2: Second Law (Entropy Generation)

$$\dot{S}_{gen} = \dot{m}(\hat{S}_2 - \hat{S}_1) - \frac{\dot{Q}}{T}$$

Since $\dot{Q} = 0$:

$$\dot{S}_{gen} = \dot{m}\Delta \hat{S}$$

For an ideal gas:

$$\Delta \hat{S} = C_P \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right) \quad (3)$$

Plugging in values ($R = 8.314 \text{ J/mol K}$):

1. **Temperature Term:** $29.1 \ln(600/290) = 29.1(0.727) = 21.16 \text{ J/mol K}$. **Note the use of absolute temperature (K) in the log term!**
2. **Pressure Term:** $8.314 \ln(10/1) = 8.314(2.302) = 19.14 \text{ J/mol K}$

$$\dot{S}_{gen} = 1 \cdot (21.16 - 19.14) = +2.02 \frac{\text{J}}{\text{K} \cdot \text{s}}$$

Since $\dot{S}_{gen} > 0$, this process is **Irreversible** and possible.

3.3 Isentropic Efficiency

We can compare the *actual* work to the *ideal* (reversible) work. We call this the isentropic efficiency of the compressor:

$$\eta_{comp} = \frac{\dot{W}_{ideal}}{\dot{W}_{actual}}$$

This will be less than 1 for a real compressor due to irreversibilities. It is often given as a percentage and is somewhere between 70-90% for typical compressors.

If the process were reversible ($\Delta S = 0$), the outlet temperature T'_2 would be (from Eq. 3 with $\Delta S = 0$):

$$T'_2 = T_1 \left(\frac{P_2}{P_1} \right)^{R/C_P} = 290(10)^{2/7} \approx 560 \text{ K}$$

The actual compressor heats the gas to 600 K, requiring more work than the ideal case (560 K). The irreversibility shows up as a higher outlet temperature and more work input.